

XINYI (CINDY) CAO

Technical Artist ♦ Tools, Pipeline & Real-Time Rendering

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SKILLS

Programming & Shaders: HLSL / ShaderLab, Python (working knowledge)

DCC, Engines & Pipeline: Proprietary DCC Tooling, Unreal Engine, Unity, PyQt, OpenUSD, Rez, Git, Linux

Technical Art: Artist Tools, Pipeline Validation, Real-Time Rendering, Profiling & Optimization, Virtual Production

WORK EXPERIENCE

LIGHTING TECHNICAL ASSISTANT at NETFLIX ANIMATION STUDIOS

Vancouver, 12/2025 to Present

- Supported lighting artists on the animated feature *Steps*, **troubleshooting** production software and workflows, resolving asset-version mismatches, investigating unfamiliar technical issues, and **diagnosing and optimizing** long-running renders
- Triaged** issues that surfaced in Lighting, identifying whether root causes lay in lighting, the renderer, the render farm, or upstream asset/animation/FX workflows, and routing them to the appropriate owners
- Developed and shipped a **contextual drag-and-drop** workflow in **Python/PyQt** within a proprietary lighting DCC, enabling artists to choose from multiple node-graph operations instead of a single fixed action
- Built and shipped **two validators** in the proprietary lighting DCC to detect performance-heavy expressions and stale scene data, including a **one-click fix** for unnecessary cached content

VIRTUAL TECHNOLOGY STAFF at CHINA EDUCATION TELEVISION (CETV)

Beijing, 07/2021 to 08/2023

- Served as the **technical point of contact** during the buildout of a proprietary virtual studio system, coordinating with external contractors from initial setup through production deployment
- Supported artists and operators using an **Unreal Engine 4 virtual studio system**, troubleshooting real-time content, software, and workflow issues under live-broadcast constraints
- Standardized a **content workflow** (import presets, naming) and 1-page checklists to reduce last-minute fixes during shows

TECHNICAL ARTIST INTERN at PERFECT WORLD

Beijing, 04/2019 to 12/2020

- Profiled and optimized rendering performance** for a mobile MMO across target devices, identifying bottlenecks, refining complex shaders, implementing Level-of-Detail (LOD) systems, and optimizing assets
- Developed and implemented **30+ shader effects** and utilities in Unity using **ShaderLab/HLSL**, supporting visual and production needs for the art team
- Prototyped new visual features and resolved **10+ technical issues** for artists and designers

SELECTED PROJECTS

REAL-TIME CHINESE INK RENDERING FOR A VR EXPERIENCE | Unity, ShaderLab/HLSL, VR

- Designed and implemented a real-time Chinese ink **NPR** system in Unity using **ShaderLab/HLSL**, combining view-dependent and shell-based contour rendering, Half-Lambert shading, diffuse warping, and triplanar brush-texture mapping for characters and environments
- Integrated and refined the rendering system for VR, resolving wide-FOV contour and mesh-dependent artefacts and validating **real-time performance** on Oculus Rift S with Oculus Debug Tool and Unity UPR

HARVEST DAY | Unreal Engine 5, Sequencer, MetaHuman, Character Pipeline

- Built a reusable **UE5** material and content workflow for a multi-environment cinematic, creating custom material effects and iterating on real-time lighting and post-processing to establish a cohesive visual style
- Developed an end-to-end **MetaHuman** character workflow, **retargeting** Mixamo motion through bone-chain and pose remapping, capturing facial performance with Live Link Face and Take Recorder, and refining animation with Control Rig

EDUCATION

Simon Fraser University: Master of Digital Media (M.DM.), Centre for Digital Media, Feb 2025

Beijing Film Academy: Master of Arts (M.A.) in Film Studies, Department of Film Technology, June 2021

Beijing University of Posts and Telecommunications: Bachelor of Engineering (B.Eng.) in Digital Media Technology, June 2018